Machine Learning-Based Early Warning System for Predicting Urban Waterlogging Risk During Heavy Rainfall

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***Abstract*—Urban waterlogging during intense or prolonged rainfall can disrupt transportation, damage infrastructure, and threaten public safety. This paper presents an experimental machine-learning prototype that examines whether rainfall patterns are associated with recorded flood events at the district-day level. Daily gridded rainfall data for 2023 from the India Meteorological Department (IMD) were processed into district-level features and matched, where possible, with a historical flood-event inventory. Six rainfall features were used to train a Random Forest classifier. A chronological split separated records before 1 August 2023 for training from records on or after that date for testing. At an experimental decision threshold of 0.01, the model achieved 94.26% accuracy, 0.47% precision, 28.57% recall, and a 0.93% F1-score. The confusion matrix reveals a large number of false alerts and missed recorded events. The high accuracy is therefore misleading under severe class imbalance. Incomplete event matching and the absence of time-aligned urban waterlogging observations limit the conclusions. The main contribution is an end-to-end academic prototype integrating data preparation, feature engineering, chronological evaluation, and a Streamlit interface; it is not a validated operational warning system.**

***Index Terms*—Urban waterlogging, rainfall analysis, Random Forest, flood-event classification, class imbalance, feature engineering, Streamlit.**

I. Introduction

Heavy or prolonged rainfall can exceed the capacity of urban drainage systems and cause water accumulation on roads, in low-lying locations, and around infrastructure. The resulting disruption can affect mobility, public safety, and economic activity. Urban flood research increasingly uses machine learning (ML) alongside physical and hydrodynamic modelling. Recent reviews discuss methods ranging from Random Forest (RF), support vector machines, and neural networks to convolutional and hybrid models [1]–[4].

The performance of a data-driven flood model depends not only on the algorithm but also on the quality, spatial coverage, and temporal alignment of its input data and event labels. This project develops a rainfall-only baseline workflow using district-day records. The goal is to demonstrate data preparation, rainfall feature engineering, chronological model evaluation, and an interactive application while explicitly identifying the limitations that prevent real-world deployment.

II. Objectives and Problem Formulation

The specific objectives are: (1) process 2023 gridded rainfall data into district-level daily features; (2) match historical flood-inventory records using district and date information where possible; (3) train a Random Forest classifier using six rainfall features; (4) evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix; (5) develop a Streamlit interface for interactive inference; and (6) document label limitations and relate the approach to recent literature.

The task is formulated as binary classification of a district-day record. The positive class indicates that a flood event was matched to that district and date in the inventory. The negative class means that no event was matched; it does not prove that flooding did not occur. This distinction is important when interpreting model errors.

III. Literature Review

Zeng *et al.* review the transition from traditional numerical urban-flood simulation to deep-learning approaches and discuss trade-offs involving computational cost, data requirements, and predictive capability [1]. Anik *et al.* examine physical-process and ML approaches to urban flood prediction, highlighting data scarcity and the value of sensor and community observations [2]. Li *et al.* review AI-based urban-flood prediction and identify commonly used inputs such as rainfall, water depth, and elevation, as well as outputs such as inundation depth and area [3].

Asif *et al.* review ML applications for short-term flood forecasting and show that model performance depends on forecast horizon, data, and evaluation protocol [4]. Islam *et al.* survey urban flood susceptibility mapping using remote sensing and ML, underlining the importance of spatial information [5]. Recent 2026 reviews further emphasize the growing range of data sources, data integration, and geospatially informed ML methods for flood prediction [6], [7]. These studies motivate the use of a simple tabular baseline for the present project, while also showing why rainfall-only features are insufficient to represent the complete physical process of urban inundation.

The studies above use different locations, target definitions, temporal resolutions, and evaluation protocols. Their published scores are therefore not directly comparable with the results in this paper. A fair numerical comparison would require

training candidate algorithms on the same project dataset and chronological split.

IV. Datasets and Preprocessing

A. Data Sources

The rainfall input is daily gridded rainfall data for 2023 from the India Meteorological Department. The source grid has dimensions of 135×129 cells and covers 365 days. The historical flood inventory used in the project contains 6,876 records spanning 1967–2023, including 634 records for 2023. After rainfall processing, the district-day feature table contains 212,064 rows across 576 districts.

The 2023 event-matching analysis identified 299 matched district-day event rows, while 142 event district-days remained unmatched. These counts describe the matching process and should not be interpreted as a complete census of all flood events. A Chennai inundation KML containing 192 locations was inspected, but reliable timestamps and verified depth units were unavailable. It was therefore not used as time-aligned supervised ground truth.

B. Feature Engineering

Six rainfall features were created: `Mean_Rainfall_mm`, `Max_Rainfall_mm`, `Rainfall_Previous_3_Days_mm`, `Rainfall_Previous_7_Days_mm`, `Rainfall_3_Day_Total_mm`, and `Rainfall_7_Day_Total_mm`. Mean rainfall represents average conditions, while maximum rainfall captures a local peak. Previous 3-day and 7-day rainfall describe rainfall history before the current day. The 3-day and 7-day totals include the current day and represent short- and longer-window accumulation. The windows were selected as practical initial features to provide the model with both recent and accumulated rainfall context.

Rows with missing feature values were removed during preparation. Fig. 1 compares average feature values for district-day records with and without a matched event. The matched-event group has higher averages across the displayed features. This is a descriptive association in the prepared dataset, not evidence that rainfall alone caused the events.

V. Proposed Architecture and Implementation

The implemented pipeline is shown in Fig. 2. It consists of rainfall and flood-inventory inputs, cleaning and district/date matching, feature engineering, chronological train/test separation, Random Forest training and evaluation, and a saved model used by the Streamlit application.

The classifier uses 200 decision trees, balanced class weights, `random_state=42`, and `n_jobs=-1`. Random Forest was selected as a practical baseline because it combines multiple decision trees, can model nonlinear relationships, and works with numerical tabular features without requiring extensive feature scaling. Balanced class weights were used to address the rarity of positive labels. This selection does not imply

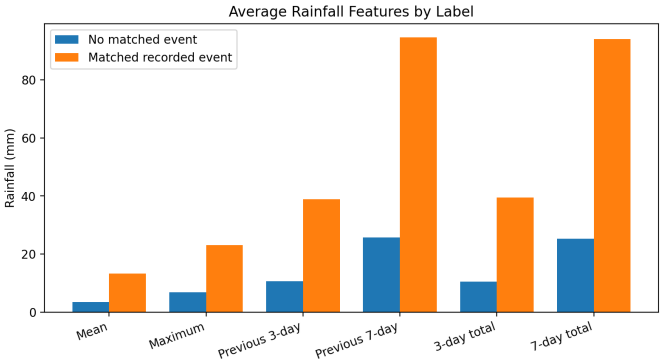


Fig. 1. Average rainfall features by recorded-event label.



Fig. 2. Architecture of the implemented waterlogging-risk prototype.

that Random Forest is universally superior to other ML or hydrodynamic approaches.

The dataset was split chronologically. Records dated before 1 August 2023 were used for training, while records dated on or after 1 August 2023 were used for testing. This preserves the time order and avoids randomly mixing earlier and later dates. It does not replace validation on independent years or cities. The saved model, `flood_risk_model.pkl`, is loaded by `app.py`. The Streamlit interface accepts the six rainfall measurements, calls the model's `predict_proba` function, and flags an input when its score crosses the experimental threshold of 0.01. The score is not calibrated and must not be interpreted as the real-world probability of waterlogging.

VI. Results and Evaluation

The training set contained 119,105 rows, including 215 recorded-event labels. The test set contained 88,892 rows, including 84 recorded-event labels. At the selected threshold of 0.01, the model generated 5,070 alerts. The confusion matrix is shown in Fig. 3; the classification metrics are summarized in Fig. 4.

Although accuracy is 94.26%, the test set is highly imbalanced: only 84 of 88,892 records have a positive matched-event label. The model produces 5,046 false positives and 60 false negatives relative to the available labels. Precision of 0.47% indicates that very few flagged records correspond to matched recorded events, while recall of 28.57% indicates that many matched events were missed. Incomplete event matching may also mean that some apparent false positives are events missing from the inventory. Consequently, the high accuracy should not be presented as evidence of reliable operational prediction.

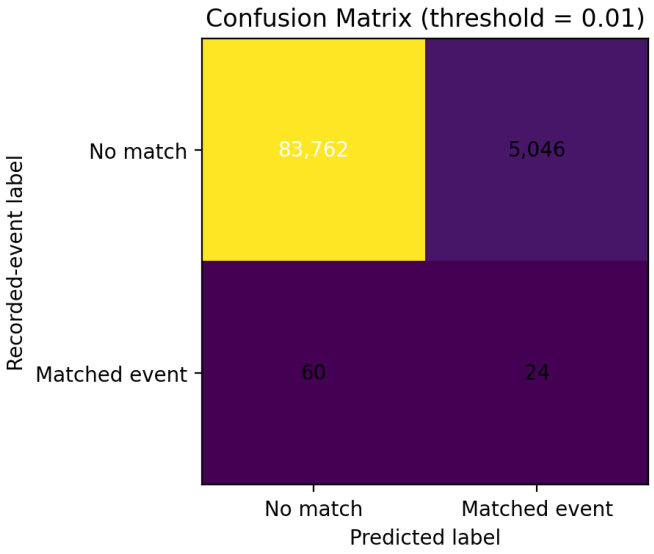


Fig. 3. Confusion matrix at threshold 0.01 (rows: recorded labels; columns: predicted labels).

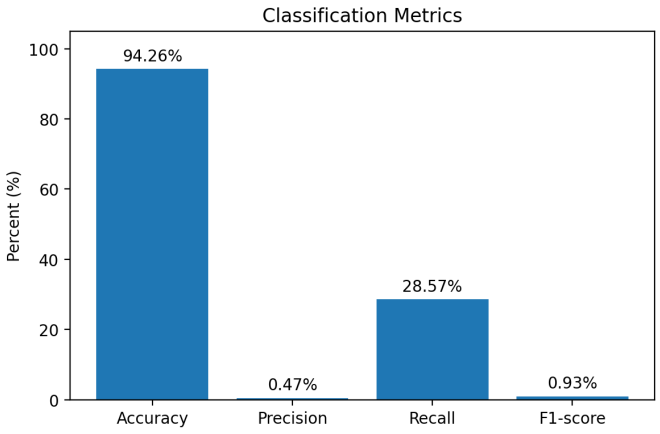


Fig. 4. Classification metrics at the experimental threshold of 0.01.

TABLE I
TEST-SET PERFORMANCE AT THRESHOLD 0.01

Metric	Value
True negatives (TN)	83,762
False positives (FP)	5,046
False negatives (FN)	60
True positives (TP)	24
Accuracy	94.26%
Precision	0.47%
Recall	28.57%
F1-score	0.93%

VII. Outcome Mapping, Novelty, and Discussion

Table II maps the project objectives to the implemented evidence. The main outcome is a complete academic prototype that connects rainfall data preparation, feature engineering, classification, evaluation, and a web interface. The contribution

TABLE II
MAPPING OF OBJECTIVES TO PROJECT OUTCOMES

Objective	Evidence / Outcome
Build district-day features	Six rainfall features; 212,064 district-day rows
Train a classifier	Random Forest model with 200 trees; saved model file
Evaluate performance	Confusion matrix and four classification metrics
Provide an interactive demonstration	Streamlit input and prediction interface
Assess practical readiness	Label limitations and validation needs documented

is application-level rather than algorithmic: the project integrates district-level rainfall features, historical-event matching, chronological testing, explicit disclosure of label limitations, and interactive inference in one workflow. No new Random Forest algorithm is claimed.

A numerical comparison against Logistic Regression, SVM, XGBoost, LSTM, CNN, or hydrodynamic models has not been performed on this same dataset. Literature reviews describe these model families, but their published results use different data and evaluation settings. Future experiments should compare candidate models on the identical chronological split and report precision, recall, F1-score, precision-recall AUC, and calibration.

VIII. Limitations and Future Work

The current experiment has several limitations: (1) event labels are incomplete and depend on district/date matching; (2) district names and historical boundaries may be inconsistent; (3) only 2023 rainfall data was used for model development; (4) no independent city- or year-based validation was performed; (5) the Chennai inundation points lack reliable timestamps and verified depth units; and (6) elevation, drainage capacity, land cover, soil, tides, and infrastructure are not included as model inputs. The score is not calibrated and should not be treated as an actual waterlogging probability.

Future work should collect verified, timestamped urban waterlogging observations, improve district/date matching, add terrain and urban drainage variables, extend the dataset to multiple years, and compare several algorithms under the same evaluation protocol. Threshold selection, uncertainty estimation, and probability calibration should also be investigated before any operational use is considered.

IX. Conclusion

This paper presented an experimental rainfall-based ML prototype for studying district-day associations with recorded flood events. The workflow processes IMD gridded rainfall, creates six district-level features, matches historical flood records where possible, trains a Random Forest classifier using a chronological split, and exposes the saved model through a Streamlit interface. The results demonstrate the end-to-end implementation, but the very low precision and incomplete ground truth prevent claims of reliable urban waterlogging

prediction. The principal outcome is a transparent baseline that identifies the data and validation improvements required for future research.

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